FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECR1213-02 NETWORK COMMUNICATIONS

LECTURER: DR. ISMAIL FAUZI BIN ISNIN

GROUP NAME: Y-LINK

NAME	MATRIC NUMBER
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Table of Content

1.0 Introduction

This project includes the design of a secure, efficient and flexible network system that will support advanced lab facilities and versatile learning spaces of the new building of the Faculty of Computing. In an interview with Mr Mohamad Nazri, he provided valuable insights into the technical and economic considerations for this development. His input helped shape a clear vision for a future-ready building that balances current needs with long-term scalability. This report will cover the interview and research findings as well as feasibility assessment to guide the project's successful implementation.

Interview Questions and Answers

1. What are the current tools requirements for each lab in the building?
   Answer: Each lab in the building requires a network switch to connect its workstations efficiently. Switches come in different port configurations, usually with 24 or 48 ports. Given that the number of workstations in each lab generally exceeds 24, opting for 48-port switches is recommended. This setup ensures seamless connectivity for all devices in each lab without needing additional switches, which optimizes space and simplifies network management. The 48-port configuration also provides flexibility for future expansion, accommodating any increase in connected devices over time.

2. Is it important to have network monitoring?
   Answer: Yes, network monitoring is crucial for maintaining the health and efficiency of a network. It provides real-time insights into the performance and status of network devices, such as switches, routers, and servers, allowing administrators to detect and resolve issues before they escalate into larger problems. Monitoring tools help identify potential bottlenecks, unauthorized access attempts, or device failures, which, if left unchecked, could lead to network slowdowns or downtime.

3. What cooling requirements should be considered for the networking equipment?
   Answer: When considering cooling requirements for networking equipment, it's essential to account for the materials used in each lab. If the lab materials are not sensitive to temperature, additional cooling measures may not be necessary. However, if the materials are temperature-sensitive, cooling systems such as air conditioning are essential to protect the equipment and maintain optimal lab conditions. Adequate cooling prevents overheating, prolongs the lifespan of networking equipment, and safeguards sensitive materials, ensuring a stable and secure environment for all lab operations.

4. Is it important to have a server room within a building ?
   Answer : Yes it is very important. Each room's switch will connect to a main switch, which is linked to a distributor switch. Each faculty will have one main switch, which will connect to a distributor switch and core switch. From there, it will connect to other switches in rooms throughout the building. First of all, it makes it possible to store crucial

data and information securely. High security protection and upkeep are built into the server room's design to guarantee the security of important data (Febryan, Purwanto, Utami, & Pramono, 2018).

5. What is the most suitable network topology for local area network (LAN) implemented in a faculty building ?
   Answer : Network topology describes the physical and logical connections made between the network's nodes. The star topology is the best network topology to use in our building. Star topology is said to be more secure and reliable because each node is separately connected to a single central hub which if one node went down, the other nodes would not be impacted. In addition, we may simply add or remove nodes as needed without impacting the network as a whole.

6. How to reduce latency in the network ?
   Answer : Delays in communication across a network are referred to as network latency. Numerous factors, including distance, transmission media, routers, and physical problems, are to blame. Network latency can be reduced in a few ways. Reduce the physical distance between the source and destination of the data first. This is due to the fact that the server can receive and react to requests more quickly when there is a shorter distance between it and the client. Next, to optimize the possible speed of the signal transmission, utilize high-quality fiber optic cable. When numerous devices connect to the network, routers with sophisticated Quality of Service (Qos) are helpful in prioritizing some traffic over others (McNally, 2021).

7. What bandwidth is needed to handle heavy usage in labs and conference rooms during peak times without slowing down?
   Answer: To ensure seamless operation and prevent slowdowns during peak usage times lab and conference rooms, it’s essential to have adequate bandwidth capacity. This can be achieved by enhancing the switch requirements to support higher internet speeds on the backend. Additionally, the connection pathway from the lab switch to the main network switch should be upgraded to a higher bandwidth, such as 4Gbps. This will allow data to flow more efficiently, accommodation heavy usage without compromising performance.

8. What backup and redundancy systems are required to keep the network reliable if hardware fails or there's a power outage?
   Answer: Reliability is crucial in a faculty where students and researchers depend on continuous network access. Implementing backup switches in critical areas is crucial especially in the event of hardware failure or power outages. Important locations such as staff rooms and labs should have a secondary switch, ensuring continuous connectivity. Furthermore, these backup switches can be equipped with a battery and Uninterruptible Power Supply (UPS) system. The UPS system serves as a buffer against power disruptions by providing a stable and uninterrupted power source in the event of

unforeseen events such as utility power outages (Neelamraju et. al, 2023). This setup allows switches to keep running temporarily during those events.

9. What firewalls and intrusion detection systems should be used to protect the network from cyber threats?
   Answer: A firewall is the primary level of protection for any system or network (Nassoura, 2022). Installing firewalls and intrusion detection systems like Avira will help protect the network against cyber threats by detecting unauthorized access attempts and blocking malicious activity. This cybersecurity measure is vital for safeguarding sensitive information and ensuring that all network users remain protected from potential attacks.

10. How many network access points will be needed per floor to ensure sufficient coverage?
    Answer: A network access point is a device that is connected to a central wired router or switch that creates a Wireless Local Area Network (WLAN) which allows devices to connect to a network such as the Internet (Anusha, 2024). Based on the interview, Mr Mohamad Nazri stated that it is important to know the number of students that the building will accommodate before deciding on the number of network access points. An access point can connect up to 200 devices simultaneously. An estimation of 500 devices can be used simultaneously on the first floor and an estimation of 210 devices will be used simultaneously at ground floor. Thus, a minimum of 3 access points on the first floor and a minimum of 2 access points on the ground floor is required. To have a better Internet connection, more network access points can be added.

11. What cabling infrastructure is appropriate for all rooms and labs?
    Answer: Mr Nazri said that twisted pair cables are optimal to be used in all rooms and labs. Cat 5, Cat 5e, Cat 6 and Cat 6a are best to be used in all rooms and labs. Cat 5 cables and Cat 6 cables have high-speed data transmission. Cat 5 cables transfer data with a rate up to 100 Mbps and has a bandwidth of 100 MHz whereas Cat 6 cables transfer data with a rate of up to 10 Gbps and a bandwidth of 250 MHz. Cat 5 cables are used to link computers on a Local Area Network (LAN) and network devices like modems and routers whereas Cat 6 cables are used when significant data transfer is required (Rich, 2024).

12. Are there guidelines or accountability criteria that the network must follow?
    Answer: Adhering to relevant regulations and standards is crucial for guaranteeing data security and accessibility. If the faculty handles personally identifiable information (PII) of EU individuals, they must comply with the General Data Protection Regulations (GDPR), which include data minimization, explicit user consent, and limited access to sensitive data. By giving consumers control over their data and minimizing needless data storage and access, GDPR compliance guarantees the protection of privacy rights (European Council, 2023).

2.0 Feasibility

1. Technical Feasibility

The proposed network system for the faculty building is technically feasible based on the infrastructure and requirements outlined. Each lab requires a 48-port switch to handle the number of workstations, ensuring seamless connectivity and room for expansion. To maintain network health and minimize downtime, network monitoring tools are necessary to provide real-time insights and issue alerts, while cooling systems such as air conditioning may be needed in labs containing temperature-sensitive materials. The design includes a dedicated server room for data centralization and security, with each room's switch connected to a main switch, distributor, and core switch for structured connectivity.

Additionally, the selected star topology is dependable and manageable, which makes it a good fit for the faculty building. Appropriate bandwidth will be allocated in labs and conference rooms to accommodate high demand, and redundancy strategies will be in place to lessen the impact of hardware failures or power outages. Each floor will have thorough coverage and speed thanks to access points and cabling, which will improve network performance overall. The technical design supports dependable, secure, and effective network operations by taking into account both present demands and possible future expansion.

2. Economic Feasibility

To assess the economic feasibility of the new building for faculty, we considered the costs associated with equipment, installation and ongoing maintenance as well as potential cost-saving measures. The proposed network design is economically feasible with necessary investments in switches, cabling and centralized server room. This will enhance the performance and reduce the future maintenance costs. The star topology simplifies troubleshooting which helps lower long-term expenses. Hence, this makes this design a cost-effective solution for the faculty's current and future networking needs.

3. Operational Feasibility

The proposed network system for the new building of the faculty is operationally feasible. This is because if the faculty anticipates a 15% growth, thus a total of 300 new students and staff are expected. The floor plan has 2 staff rooms which accommodates a total of 52 staff and has multiple labs and a hybrid classroom which all simultaneously can accommodate a total of approximately 180 students. Hence, a total of 230 people can be accommodated at once in this new building. Since the faculty of computing does not solely depend on this building and has other two buildings to accommodate students and staff, this proposed floor plan and network system is appropriate, reasonable and efficient. The new building will be in use as all four labs on the first floor which are two general labs, one CISCO lab and one embedded lab will be used for teaching. As for the ground floor, the student lounge will definitely be in use as students will rest or do their work there while waiting for their next classes. WiFi connection in the student lounge will further promote its use. Besides that, the hybrid classroom can be used for classes that require students to be more interactive and the conference room can be used by staff and also students if they have industrial talk. Overall, the proposed floor plan and network system for the new building is operationally feasible.

4. Security Considerations :

Security considerations are a critical component of the proposed building project, particularly regarding network safety to safeguard data. Measures such as firewalls and antivirus software are proposed to protect against potential threats. Balancing security with convenient access for users is also a priority, ensuring that data is well-protected. These strategies aim to create a secure environment suitable for academic activities.

3.0 Minute meeting

Date/Time		5 Nov 2024, 8.45 p.m	
Location		Online (Whatsapp)	
Agenda		1. Discuss the questions on Task 1	
		2. Decide the interviewee	
		3. Task distribution for each person	
Meeting Host		Safiya Nursyahadah	
Attendance			
Name		Time	Reason For Absence
Dayang Farah Farzana		8.45 p.m	-
Farra Nurzahin		8.45 p.m	-
Harini		8.45 p.m	-
Safiya Nursyahadah		8.45 p.m	-
Minutes			
No.	Item Discussed	Details	Person in charge
1.	A short discussion on question given on Task 2	- All members read the question and rubric together.	All members
2.	Decide the interviewee	- Each member suggested a technician of UTM's faculty of computing to be interviewed for Task 2. - Safiya suggested Mr Mohd Idzam Iqbal bin Abd Rani to be interviewed. - Harini suggested to	All members

		interview Mr Mohmad Azmi bin Latonak @ Ismail. - Farra suggested to interview Mr Mohmad Azmi bin Latonak @ Ismail and Miss Norhani binti Mohd. Haitami. - Dayang suggested to interview Mr Mohmad Azmi bin Latonak @ Ismail and Mr Mohamad Nazri bin Samin.	
3.	Email the interviewees	- Farra and Dayang emailed the interviewees about the purpose of the interview and asked the availability of the interviewees to book a slot for the interview.	Farra, Dayang
4.	Task distribution	- We distribute the tasks using the “Spin The Wheel” method. - Outcomes : Safiya is responsible for researching about the security considerations of the network plan, Dayang is responsible for researching about the technical feasibility of the network plan, Farra is responsible for researching about the economical feasibility of the network plan and the introduction, and	All members

		Harini is responsible for researching about the operational feasibility of the network plan. - Each members are also assigned to generate 3 questions to be asked to the interviewee.	
4.	Meeting ended	- At 9.30 p.m., Safiya ends the meeting after all discussions have been done.	Safiya

<table>
<tr><td colspan="2">Date/Time</td><td colspan="2">7 Nov 2024, 2.30 p.m.</td></tr>
<tr><td colspan="2">Location</td><td colspan="2">MPK 2, N28a</td></tr>
<tr><td colspan="2">Agenda</td><td colspan="2">Interview Mr Mohamad Nazri bin Samin about the requirements that are necessary to develop a network plan for the proposed building floor plan.</td></tr>
<tr><td colspan="2">Meeting Host</td><td colspan="2">Farra Nurzahin</td></tr>
<tr><td colspan="4">Attendance</td></tr>
<tr><td colspan="2">Name</td><td>Time</td><td>Reason For Absence</td></tr>
<tr><td colspan="2">Dayang Farah Farzana</td><td>2.30 pm</td><td>-</td></tr>
<tr><td colspan="2">Farra Nurzahin</td><td>2.30 pm</td><td>-</td></tr>
<tr><td colspan="2">Harini</td><td>2.30 pm</td><td>-</td></tr>
<tr><td colspan="2">Safiya Nursyahadah</td><td>2.30 pm</td><td>-</td></tr>
<tr><td colspan="4">Minutes</td></tr>
<tr><td>No.</td><td>Item Discussed</td><td>Details</td><td>Person in charge</td></tr>
<tr><td>1.</td><td>A short briefing on Task 1</td><td>- Harini explained why the interview was arranged for and explained about the assignment in detail.
- Harini showed the floor plan that has been designed by all members to Mr Nazri.</td><td>Harini</td></tr>
<tr><td>2.</td><td>Asking for Mr Nazri’s consent</td><td>- All members asked for Mr Nazri’s consent to record the interview session for referential purposes. Mr Nazri consented for the interview to be recorded.
- Dayang recorded the interview session.</td><td>Dayang</td></tr>
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3.	Interview session	- All members took turns to interview Mr Nazri regarding the questions that they had previously prepared. - Once all members had asked their questions and got their answers, the interview session ended.	All members
4.	Meeting ended	- Once all members had asked their questions and got their answers, the interview session ended. - All members thanked Mr Nazri for his cooperation. - Dayang stopped the recording.	All members

4.0 References

Anusha. D, “Network access points: What are they & why do you need them?,” *meter.com*, para. 1, August 1, 2024. [Online]. Available: https://www.meter.com/resources/network-access-points. [Accessed November 8th, 2024].

European Council, “The general data protection regulation,” *www.consilium.europa.eu*, Dec. 11, 2023. [Online]. Available: https://www.consilium.europa.eu/en/policies/data-protection/data-protection-regulation/. [Accessed November 8th, 2024].

McNally, C. (2021, October 15). What is latency and how do you fix it? Reviews.org. Retrieved November 15, 2021, from https://www.reviews.org/internet-service/what-is-internet-latency/

Nassoura, A. B, “Cybersecurity Technologies And Practices In Higher Education Institutions: A Systematic Review”. *Webology*, *19*(3). 2022. [Online]. Available: https://www.researchgate.net/profile/Ayman-Nassuora-2/publication/360835004_Cybersecurity_Technologies_And_Practices_In_Higher_Education_Institutions_A_Systematic_Review/links/628dcf816773462154d7116e/Cybersecurity-Technologies-And-Practices-In-Higher-Education-Institutions-A-Systematic-Review.pdf [Accessed November 9th, 2024].

Neelamraju, P. M., & Yellampalli, S. (2023) “Analysis of uninterruptable power supply critical-to-quality factors.” *Journal of Power Electronics*, *23*(12), 1919-1930. May 2023. [Online]. Available: https://www.researchgate.net/profile/Pavan-Neelamraju/publication/374372112_Analysis_of_Uninterruptable_Power_Supply_Critical-to-Quality_Factors_Bachelor_of_Technology/links/651a72d9321ec5513c2a8a21/Analysis-of-Uninterruptable-Power-Supply-Critical-to-Quality-Factors-Bachelor-of-Technology.pdf [Accessed November 9th, 2024].

24. Febryan, Hari, Purwanto., Ema, Utami., Eko, Pramono. (2018). Implementation and Optimization of Server Room Temperature and Humidity Control System using Fuzzy Logic Based on Microcontroller. doi: 10.1088/1742-6596/1140/1/012050